

(parts per million). The closest standard resistance value is E192 = 3.92k Ω .

Note. Preferred numbers/values are standard guidelines used by manufacturers to choose exact product dimensions for given constraints.

Renard's numbers/values are a set of preferred values defined by the French army engineer, Colonel Charles Renard, and is used within the SI system.

E series is another set of preferred numbers proposed by international standard IEC 60063 for electronic components such as resistors, inductors, capacitors and Zener diodes.

Calculation of LED resistance

Every LED circuit should have a resistor connected in series in order to work properly. The second tab in the window, *Calculate LED resistors*, is where you can calculate the resistor value required for an LED circuit. When the user enters the supply voltage (in volts), voltage

drop across LED (in volts) and the current of the LED (in milliamperes), the value of resistor to be connected to this circuit and the power dissipation is calculated and displayed by the program. If two or more LEDs are connected in a series or parallel, the resistor value for that circuit can also be found out by entering the number of LEDs in series/parallel under its corresponding sections *LEDs in series connection/LEDs in parallel connection*.

By clicking on the *Show resistor* button, the calculated value is shown in *Resistors* tab where you can determine its colour code.

Example. Suppose you want to calculate the value of resistor for a single LED where supply voltage is 24V, drop across the LED is 3V and LED current is 25mA. Enter these values in their respective fields, and the *Resistor value* is calculated by the program as 840 Ω . Power dissipation is also calculated automatically, which is 525mW in this case.

Now consider another case where the parameters are the same but five LEDs are connected in a series. Under the subheading *LEDs in series connection*, enter the corresponding values. So the *Resistor value* and *Power dissipation* will be displayed as 360 Ω and 225mW, respectively. By clicking on the *Show resistor* button, you are taken to the *Resistors* tab, where the resistance and colour coding of the calculated resistor value of LED(s) is given.

In a similar way, if you require resistor value for LEDs connected in parallel, enter the respective values under *LEDs in parallel connection* field.

Resistor Calculator is a great tool for calculating and determining the values of resistors. It has a really simple and straight-forward interface that makes it easy to use even for first-time users. In terms of size, this software occupies little space. ●

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